

Google Fiber Repeat Calls Dashboard

Executive Summary

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Project Objective: *This project builds an interactive BI dashboard for Google Fiber's customer service call center to explore trends in repeat customer calls and help leadership understand how effectively the team resolves customer issues on the first contact.*

1. Problem & Solution Overview

THE PROBLEM

Google Fiber's call center lacked a self-serve tool to track and visualize repeat customer contacts. When customers call back after their first inquiry, it signals a failure in first-call resolution—directly impacting customer satisfaction and operational efficiency.

Leadership required clear answers to several core business questions:

- How often are customers contacting support more than once about the same issue?
- Which markets and problem types are driving the most repeat calls?
- What days of the week see the highest call volumes for staffing optimization?

THE SOLUTION

A BI analyst designed and built an end-to-end dashboard solution, mapping requirements from stakeholder interviews through a unified data pipeline to a published interactive dashboard.

The resulting architecture leverages **Google BigQuery** for robust data processing and consolidation, and **Tableau Public** for dynamic, self-serve data visualization and reporting.

2. Project Framework & Implementation Stages

The development lifecycle was structured across three distinct phases to guarantee alignment with operational goals:

Stage	Activity	Output
Stage 1 - Requirements	Stakeholder interviews with the Google Fiber call center team	Stakeholder Requirements, Project Requirements, and Strategy Documents
Stage 2 - Data Pipeline	UNION ALL SQL query in BigQuery to merge 3 distinct market tables	Single consolidated target table consisting of 1,350 rows × 11 columns
Stage 3 - Dashboard	7 interconnected worksheets built in Tableau Public across 4 dashboards	Published interactive dashboard featuring comprehensive filters and intuitive navigation

3. Operational Implementation & Capabilities

Keys to Success:

- **Data Consolidation:** Data from three independent markets was consolidated using `UNION ALL`, perfectly preserving all 1,350 records with an identical schema across all sources.
- **Self-Serve Architecture:** The dashboard is fully self-serve. Leadership can seamlessly filter metrics by market, problem type, date range, and day of the week without requiring BI team intervention.
- **Comprehensive Visibility:** Utilizing 7 worksheets across 4 dedicated dashboards enables both absolute raw counts and percentage-based views for complete context.
- **Ongoing Usability:** Automated data refresh protocols ensure the dashboard stays current dynamically as new operational call data is ingested.

Dashboard Capabilities:

- **Repeats by Month:** All 8 historical contact columns are chronologically grouped by quarter and month.
- **Cross-Referenced Tables:** Raw data is cross-tabulated by market and problem type with automated column and row totals.
- **Day of Week Analysis:** Highlights percentage of calls by weekday, instantly revealing Monday as the peak operational day (~18%).
- **Market Benchmarking:** Compares Day 0 vs. Day 1 repeat rates by market, clearly isolating specific operational concerns in `market_3`.

4. Results Summary & Success Criteria Evaluation

The deployed dashboard successfully addresses and answers the primary business questions across all four views. Initial contacts (`contacts_n`) run 10-15x higher than Day 1 repeat calls, confirming that the majority of customer issues are resolved on first contact. However, the absolute volume of repeat inquiries remains significant enough to warrant targeted tactical action.

Success Criteria Requirement	Result Status
Dashboard explicitly shows how often customers make repeat calls	✓ Met – Fully visible across all 4 dashboards
Granular breakdown by market city (3 distinct markets)	✓ Met – All markets fully visible in all views
Granular breakdown by problem type (5 unique types)	✓ Met – <code>type_1</code> through <code>type_5</code> standard across all views
Call trends are viewable by week, month, quarter, and year	✓ Met – Global date filters applied on all dashboards
Completely self-serve for leadership (zero BI team dependency)	✓ Met – Interactive user-driven filters throughout
Project delivered within the strict 6-week timeline	✓ Met

5. Strategic Reflections & Next Steps

- **Optimize Staffing Allocations:** Since Monday consistently drives the highest call volume (~18% of the weekly total) across all three analyzed months, call center leadership should immediately review and scale up staffing levels on Mondays to minimize wait times and improve initial resolution.
- **Rebalance Resources by Market Volume:** `market_1` handles an overwhelmingly larger absolute call volume compared to others (45,333 initial contacts vs. only 4,389 for `market_2`). Ongoing capacity planning and budget/resource allocation must reflect this massive disparity.
- **Targeted Agent Training:** Classification categories `type_2` (technician troubleshooting) and `type_5` (internet and wifi issues) generate the highest volume of repeat calls. Designing and delivering targeted refresher training for customer service agents in these two categories is highly recommended.
- **Incorporate Resolution Outcome Data:** Future dashboard iterations should integrate definitive call resolution outcome data. This will allow the calculation of true, verified first-call resolution (FCR) rates rather than relying strictly on subsequent repeat contact counts as a proxy.
- **Scale the Framework:** The architectural framework and data pipelines established during this project are fully scalable. They can be effortlessly extended to accommodate new additional markets and future fiscal quarters as Google Fiber continues to expand its geographic footprint.